



INTE130 – Object-Oriented Programming Assignment.

(CyberShield system Project – Assignment Report.)

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The introduction:

In this project, we designed a simple system for a one-person cybersecurity company and called it the **CyberShield System. The goal of the project is to apply object-oriented programming (OOP) concepts in a practical way, by creating a system containing several agents that help the founder in executing tasks and making decisions**

Part A – Understanding the idea :

- In this part, we learned the concept of a one-person company, where one person is responsible for running the entire business. We also understood how digital systems can help reduce effort and make task management easier

Also, we were introduced to the idea of multi-agent systems, where each agent performs a specific task, and this helps to get the work done faster and more organized

Part B – Project idea:

We proposed the **CyberShield System** as a project idea, a system that provides cybersecurity services to small and medium-sized businesses.

This system helps monitor systems, detect problems, and suggest solutions.

First, we proposed several proxies, such as:

1- Financial agent

2-Threat monitoring agent

3-Vulnerability analysis agent

4-Response agent

5-Reporting agent

But after thinking, we chose to implement only three agents in the code, which are:

- 1- Threat monitoring agent.**
- 2- Vulnerability analysis agent.**
- 3- Response agent.**

This is because they represent the core functions of the system, while the rest of the agents are left as ideas for future development.

Part C – System Design:

In this part, we design the system's appearance before writing the code. We identified key components such as the founder, mission, and agents, and explained how each part interacts with the other.

We also drew a UML diagram to illustrate the relationship between these components, where the founder creates the task, then it is sent to the agents, and each agent handles it according to its function.

Part D – System Implementation:

In this part, we wrote a simple code using Python to implement the idea we designed.

The system works as follows:

- 1-The founder creates a mission.**
- 2-The task is sent to the agents.**
- 3-Each agent performs a task analysis.**
- 4-The results are displayed.**
- 5-Ultimately, the founder makes the decision.**

This part helped us understand how to convert the design into actual software.

Part E – Certificate:

We completed a programming course, which helped us improve our skills and understand the concepts better. This played a major role in our ability to implement this project.

The challenges we faced:

While working on the project, we faced some difficulties, such as:

Understand how to use classes and link them together

Ensure that the code works without errors

Distribution of work among team members.

**But through our cooperation, we were able to
.solve these problems.**

What we learned from the project:

Through this project, we learned:

1-How to use object-oriented programming in practice.

2-The importance of dividing the system into small parts.

3-How to work within a team.

4- :How to turn an idea into a simple application.

Conclusion and future development:

Overall, this project was a useful experience for us, as it helped us understand the concepts more deeply. The courses we took also helped us develop our skills.

In the future, we aspire to develop this system and add more features, such as improving analysis and making the system more realistic and powerful in ..dealing with threats.